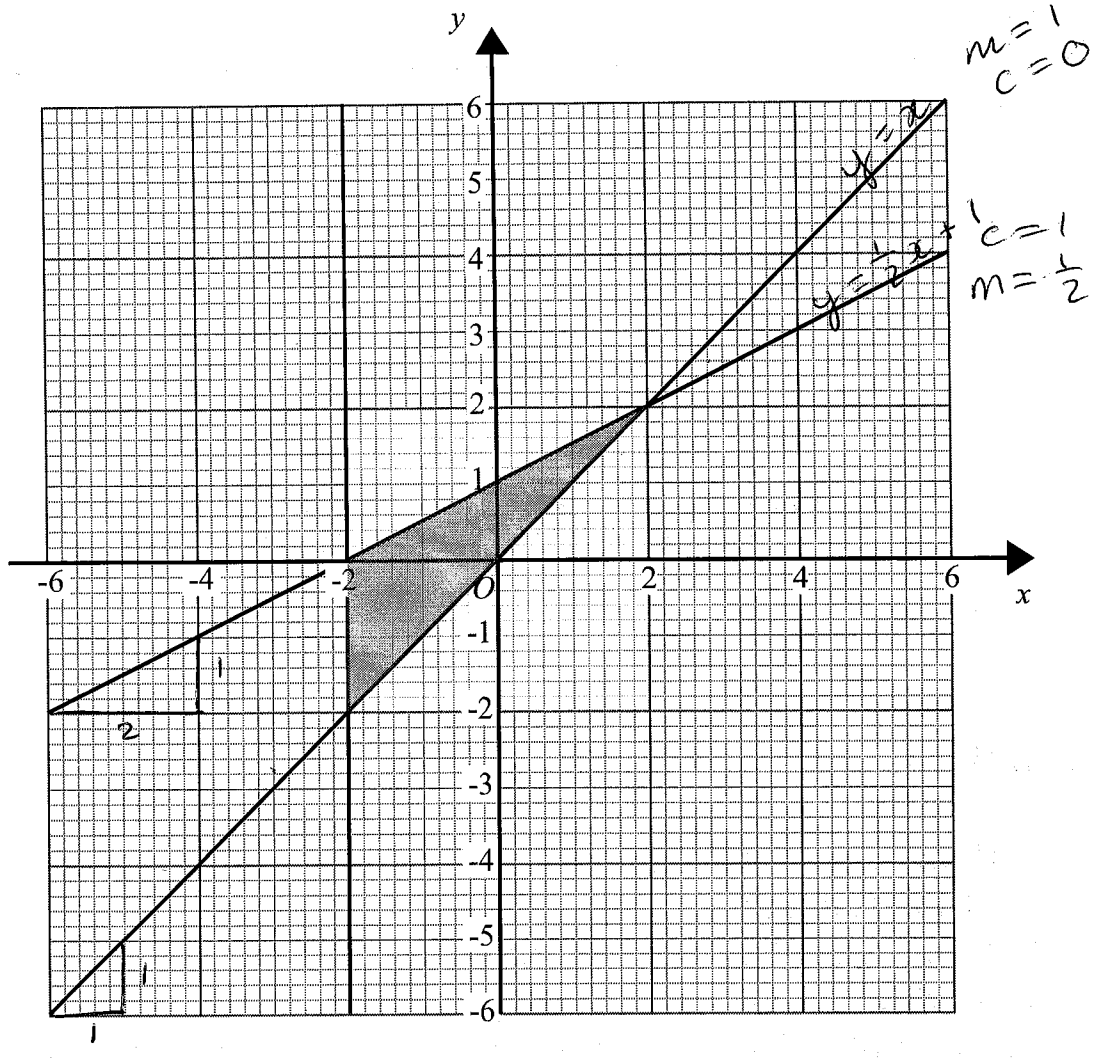
1



Write down the three inequalities that define the shaded region

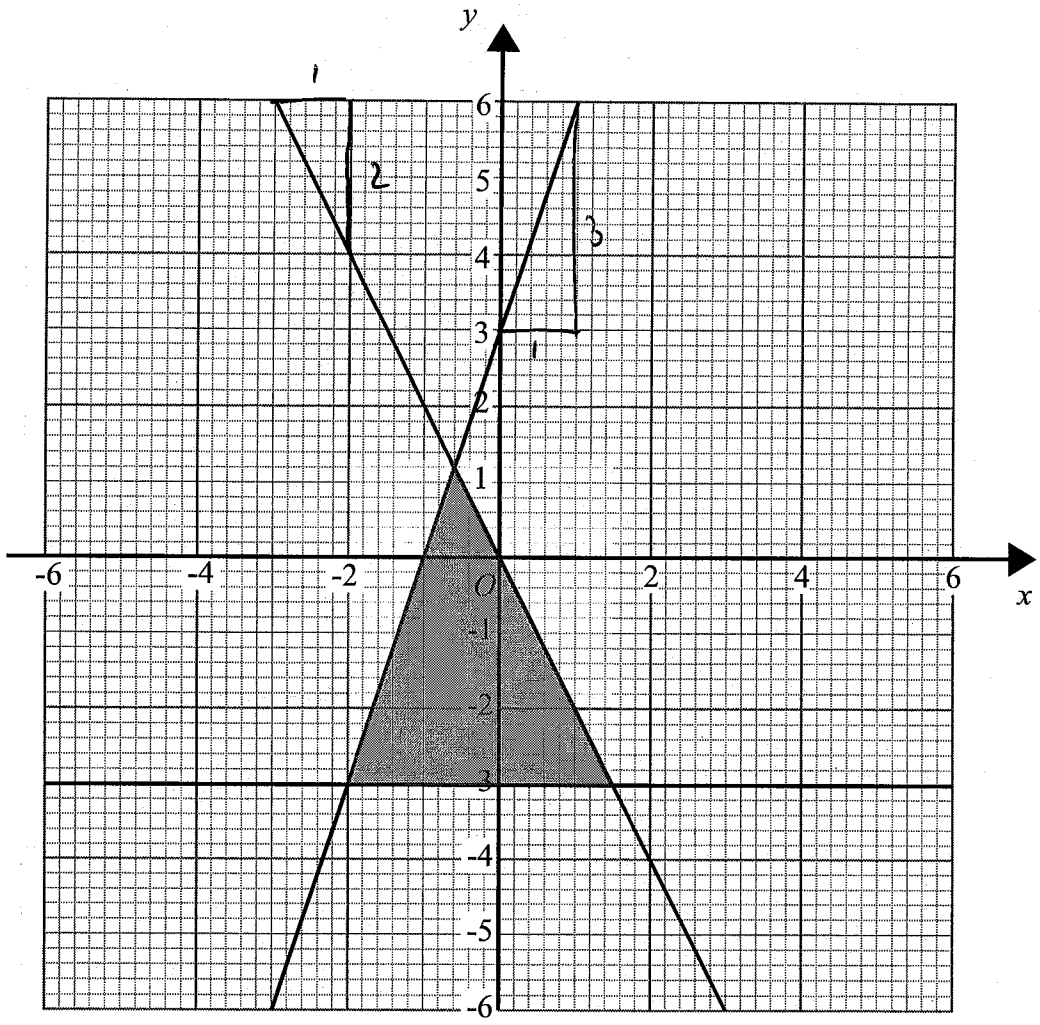
$x \geqslant -2$

$y \geqslant x$

$y \leqslant \frac{1}{2}x + 1$

(Total for Question 1 is 4 marks)

2



Write down the three inequalities that define the shaded region

$y \geqslant -3$

$m = 3$
$c = 3$
$y \leqslant 3x + 3$

$m = -2$
$c = 0$
$y \leqslant -2x$

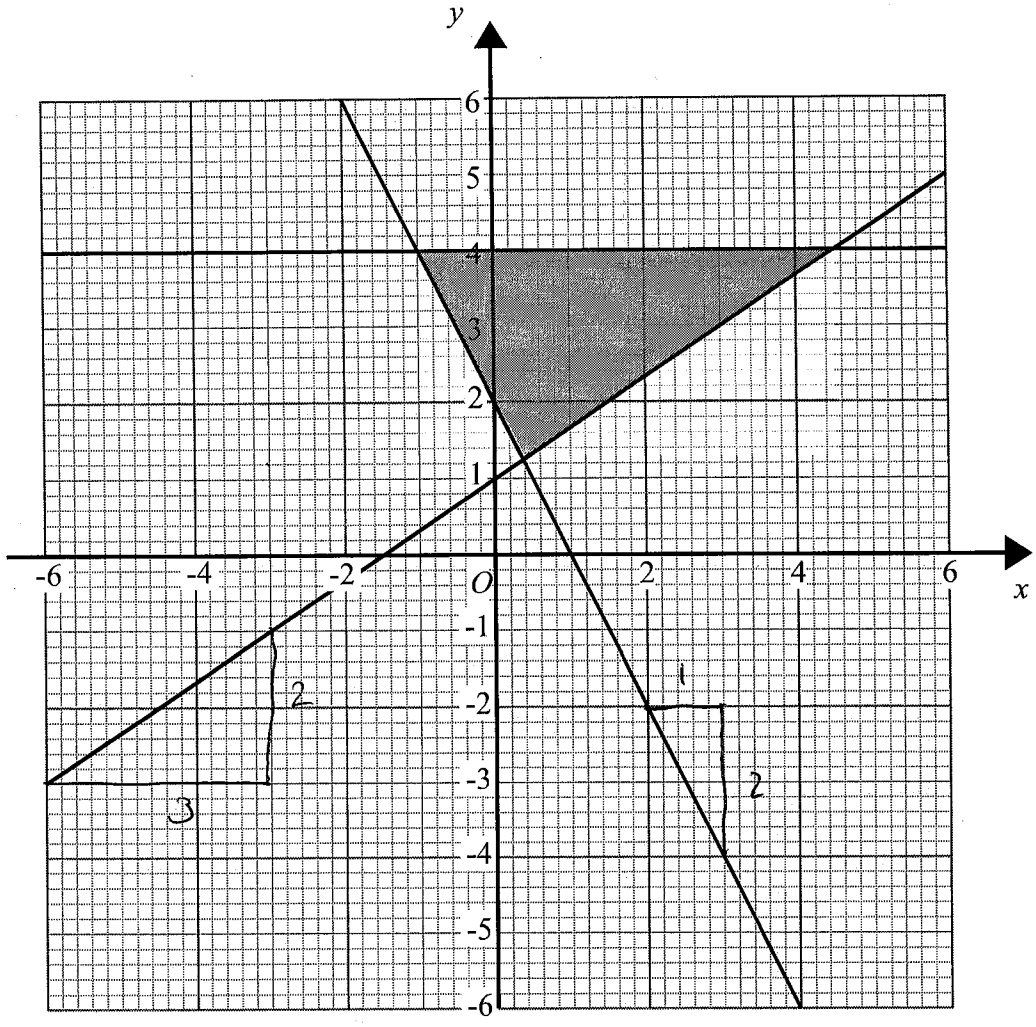
$y \geqslant -3$

$y \leqslant 3x + 3$

$y \leqslant -2x$

(Total for Question 2 is 4 marks)

3



Write down the three inequalities that define the shaded region

$y \leq 4$

$m = \frac{2}{3}$

$c = 1$

$y = \frac{2}{3}x + 1$

$y \geqslant \frac{2}{3}x + 1$

$m = -2$

$c = 2$

$y = -2x + 2$

$y \geqslant -2x + 2$

$y \leqslant 4$

$y \geqslant \frac{2}{3}x + 1$

$y \geqslant -2x + 2$

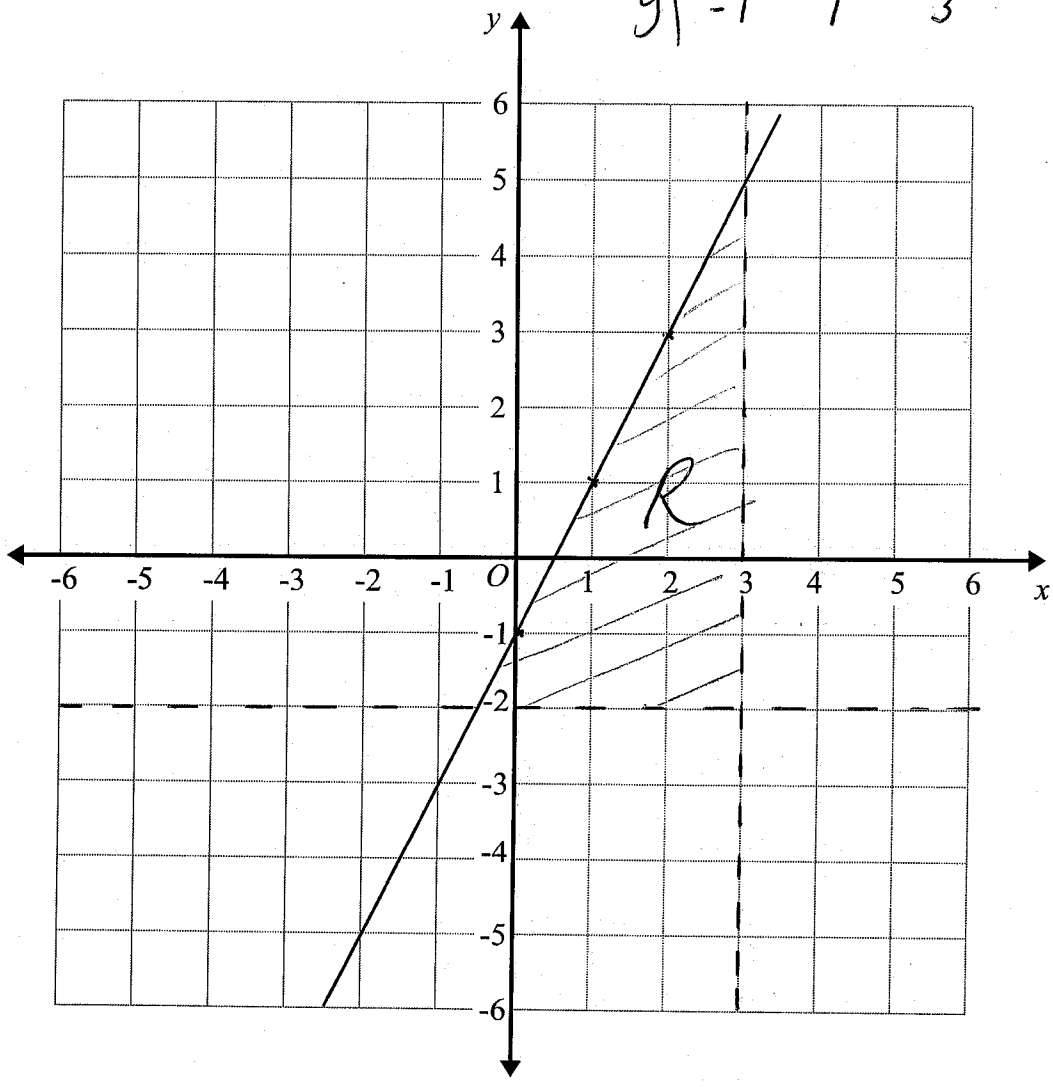
(Total for Question 3 is 4 marks)

4 On the grid shade the region that satisfies all these inequalities

$x < 3$ $\quad$ $y > -2$ $\quad$ $y \leq 2x - 1$

Label the region **R**.

x	0	1	2
y	-1	1	3



(Total for Question 4 is 3 marks)

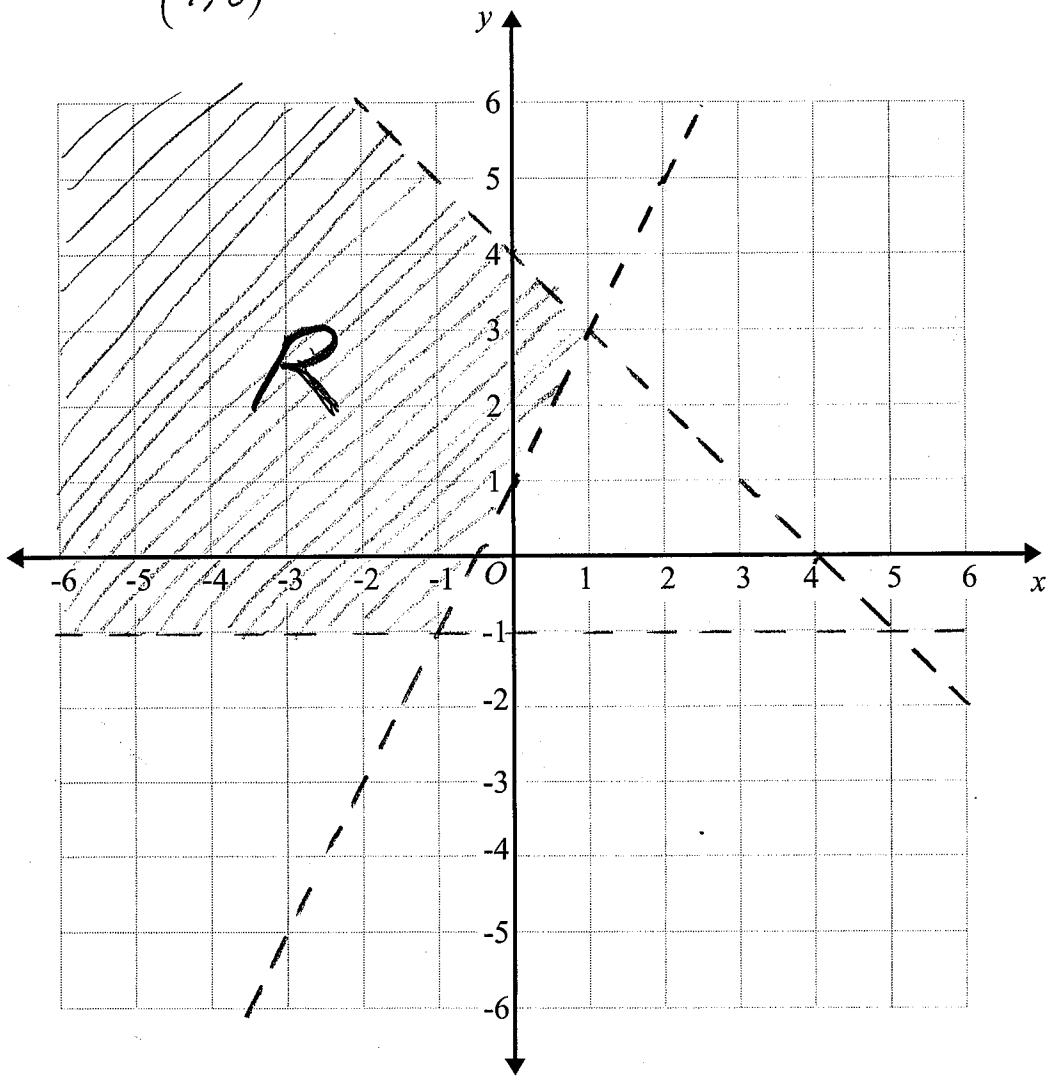
5 On the grid shade the region that satisfies all these inequalities

$x + y < 4$ $\qquad$ $y > 2x + 1$ $\qquad$ $y > -1$

Label the region **R**. (0, 4)

(4, 0)

x	0	1	2
y	1	3	5



(Total for Question 5 is 3 marks)

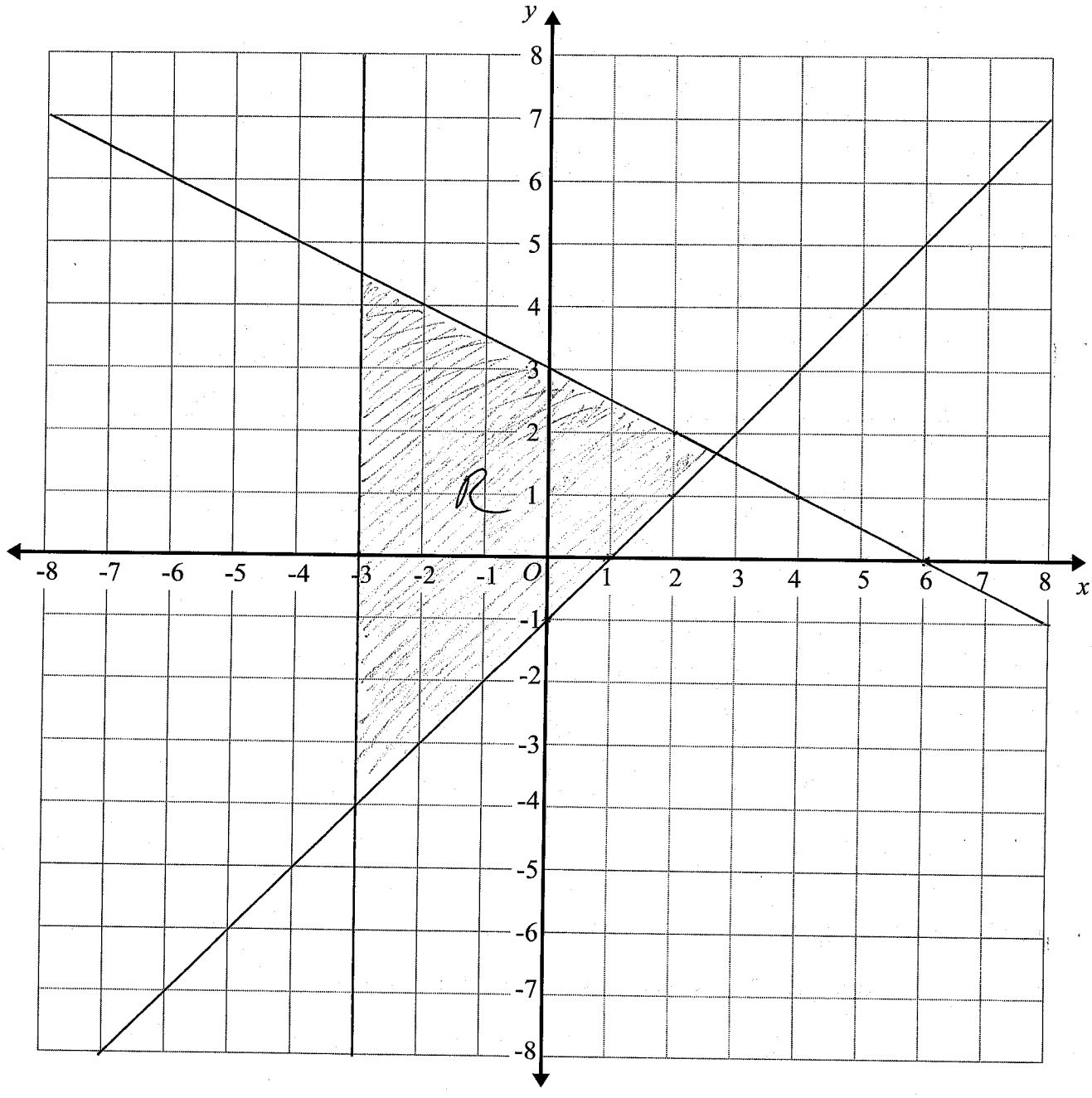
6 On the grid shade the region that satisfies all these inequalities

$y \geq x - 1$ $\qquad$ $x \leq 6 - 2y$ $\qquad$ $x \geq -3$

Label the region **R**.

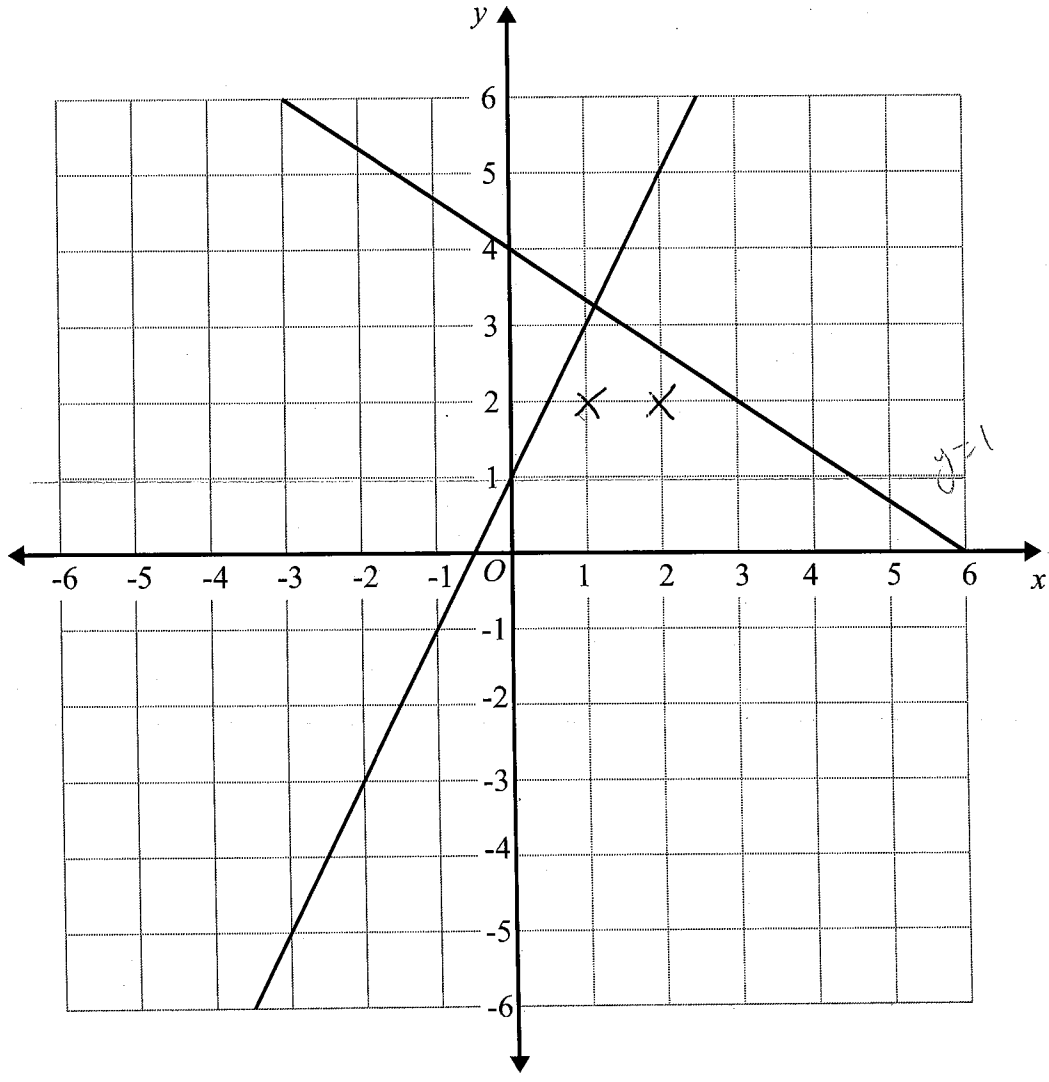
x	0	1	2
y	-1	0	1

y	0	1	2
x	6	4	2



(Total for Question 6 is 3 marks)

7 The graphs of the straight lines with equations $y = 2x + 1$ and $3y + 2x = 12$ have been drawn on the grid.



x and y are both integers.

Mark with a cross (×) all of the points that satisfies all the inequalities

$y < 2x + 1$ $\qquad$ $3y + 2x < 12$ $\qquad$ $y > 1$

(Total for Question 7 is 2 marks)